

PURPOSE

- To provide a warning system against common driving mistakes caused by temporary distractions, drowsiness, or poor visibility.
- Blindspot Monitoring: To continuously scan areas of the road that are physically blocked from the driver's view by the vehicle's pillars or bodywork.
- Relative Velocity Tracking: To measure the closing speed between SafeSight and the vehicle behind, alerting the driver if the "time-to-collision" drops below a safe threshold.
- Lane Departure Warning: to prevent "single-vehicle accidents" where a car leaves the shoulder of the road due to driver inattention or sleep apnea.

BACKGROUND

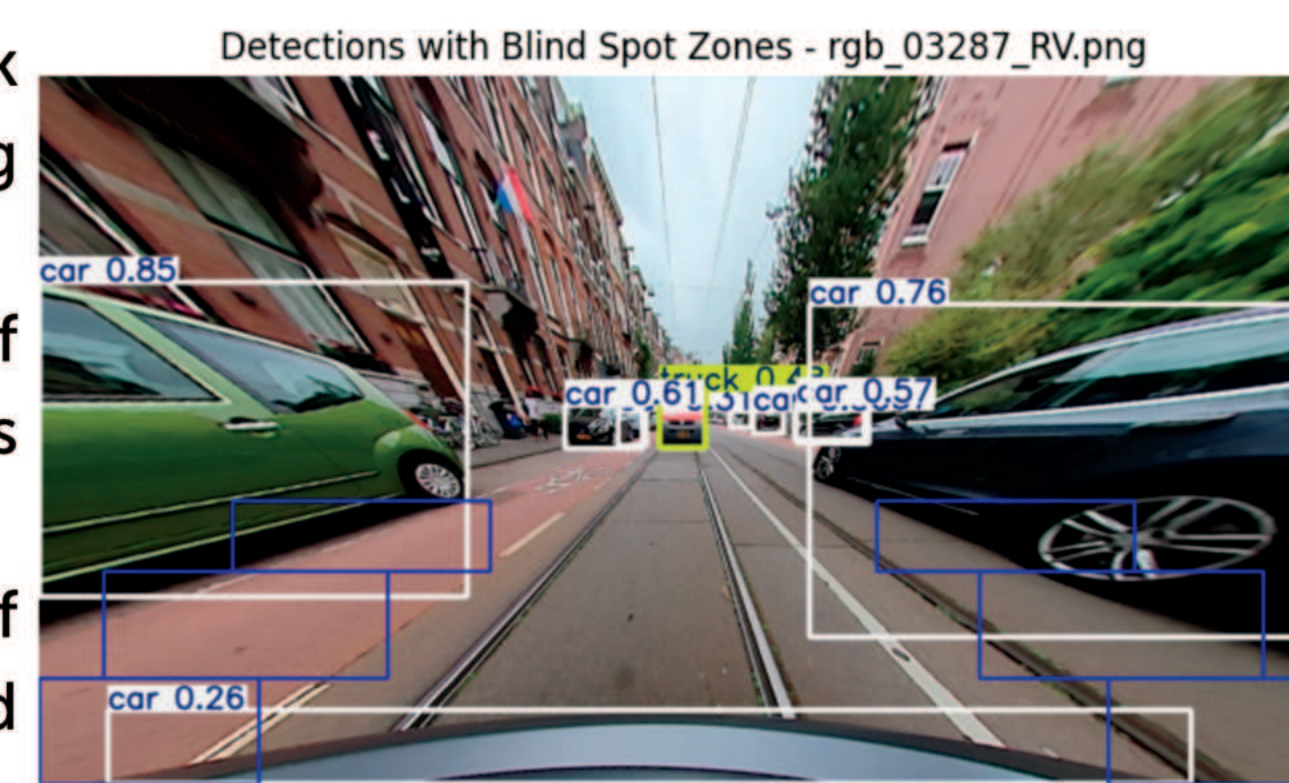
- **Hardware Availability:** High-quality Advanced Driver Assistance Systems (ADAS) are often locked behind the premium price tags of luxury vehicles. SafeSight is designed to leverage accessible hardware (like standard dashcams and affordable edge-computing devices) to bring safety technology to older or economy vehicles.
- **Human Factor Limitations:** Research indicates that nearly 94% of accidents are caused by driver behavior, such as distracted driving, fatigue, or delayed reaction times. This project bridges the gap between human perception and real-world hazards (Yazdi, 2025) .
- **Computer Vision:** With the maturity of models like YOLO (You Only Look Once) and geometric transforms like the Hough Transform, it is now possible to process complex road data in real-time on localized hardware without needing an expensive rig (Patuck, 2024) .
- Research in automotive safety proves that even an extra 0.5 to 3 seconds of warning can help tremendously to prevent rear-end collisions. SafeSight is grounded in the goal of providing this "Perceptual Buffer," giving drivers that vital extra moment to react, brake, or steer away from a developing hazard that they may not have noticed yet (Szafranski, 2024) .
- There is currently a massive "safety gap" between new vehicles equipped with factory ADAS and the hundreds of millions of older cars still on the road. The project is grounded in the necessity of retrofitting older vehicles with modern safety logic, ensuring that life-saving technology is a standard right of all drivers rather than a luxury reserved for new car buyers.

SafeSight: Driver Aid System for Blindspot Detection, Collision Warning, and Lane Departure.

METHODOLOGY

Blindspot Detection

- **Real-Time Object Detection:** Deploys the YOLOv8 (You Only Look Once) neural network to scan incoming camera frames and classify relevant traffic targets, specifically filtering for cars, motorcycles, buses, and trucks.
- **Geometric "No-Zone" Mapping:** Establishes predefined, coordinate-based Regions of Interest (ROIs) on the left and right edges of the camera feed. These virtual polygons act as the physical blind spots of the vehicle.
- **Spatial Intersection Logic:** Continuously tracks the spatial coordinates (bounding boxes) of all detected vehicles and mathematically evaluates if they intersect with the mapped blind spot zones.



Collision Warning

- Utilizes OpenCV's fisheye models to undistort the raw camera feed, specifically tuning the focal scale parameters to generate a "deep view" optimized for forward/rear distance tracking.
- Processes the frame through a Canny Edge Detector and Hough Line Transform to track left and right lane boundaries, mathematically projecting a solid "Lane Polygon" onto the road surface.
- Computes the relative distance to in-lane vehicles using pinhole camera geometry by dividing the product of the assumed real vehicle width and focal length by the bounding box pixel width.



Lane Departure

- Applies the Hough Line Transform to extract linear pixel segments within a defined trapezoidal Region of Interest (ROI).
- Implements an Exponential Moving Average (EMA) mathematical filter to carry a weighted "memory" of lane coordinates across sequential frames. This stabilizes the visual overlay, preventing the tracking polygon from collapsing or flickering when passing over faded paint or the gaps in dashed lines.
- Continuously computes the pixel deviation between the user vehicle's fixed camera center and the center of the lane, warning the user if the distance surpasses a certain threshold.

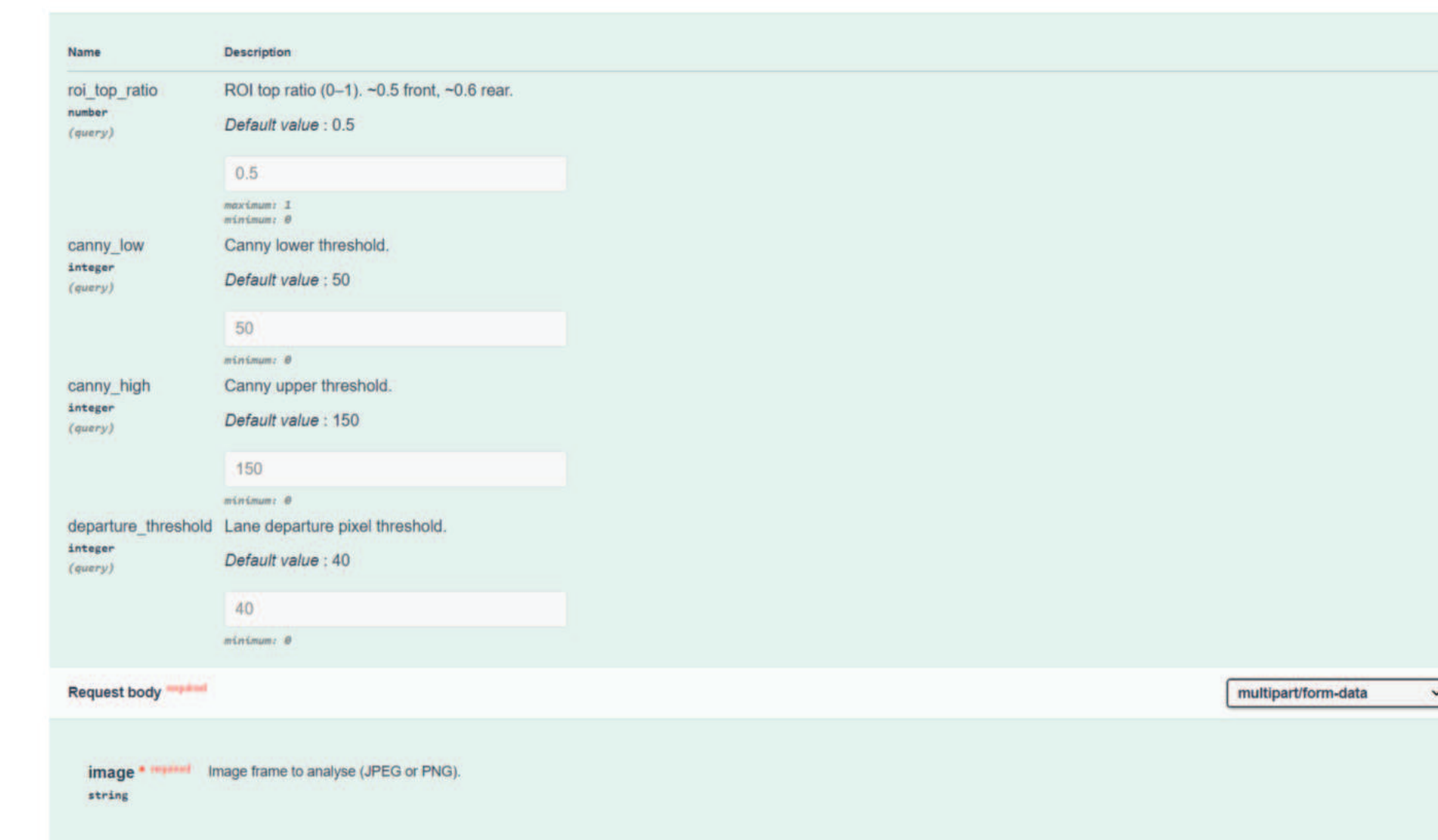


Limitations

- In order to run on a Raspberry Pi 5, the system is constrained to using YOLOv8n (the "nano" architecture). While highly optimized for speed, this lightweight model inherently sacrifices detection accuracy for distant, small, or partially occluded vehicles compared to heavier, more computationally expensive models.
- SafeSight operates purely as a vision-based system without the integration of active sensors (such as Radar or LiDAR), it is susceptible to the physical limitations of camera lenses. Performance and lane-tracking reliability degrade in adverse environmental conditions, including heavy precipitation, fog, direct solar glare, or unlit nighttime driving.
- The development of Exponential Moving Average (EMA) for lane smoothing versus tuning the fisheye distortion maps for collision warning—relied on separate, independent datasets. Because the system was not trained and validated on a single, unified, multi-camera driving dataset, its algorithms may require recalibration to be effectively deployed on a unified real-world vehicle rig.

RESULTS

- Operating entirely on a localized Raspberry Pi 5, the YOLOv8n spatial intersection logic successfully processed live video feeds for blindspot detection at a consistent 3–6 FPS, providing reliable threat alerts without needing cloud computing.
- The integration of an Exponential Moving Average (EMA) consistently performed well across all test scenarios. It effectively fixed frame-to-frame line flickering, ensuring the lane-tracking polygon remained stable even on dashed or faded road markings.
- Lastly, a website was created, shown below, to allow a user to upload their own images, allowing them to test the different systems. Also, they are able to configure the parameters to their own liking if they wish.



CONCLUSION

- Ultimately, SafeSight fulfills its primary directive: expanding the driver's spatial awareness and increasing the amount of time they have to prevent blindspot collisions, forward collisions, and unintentional lane departures.
- **Future Work:**
 - Expanding the system beyond monocular vision by integrating cheap millimeter-wave (mmWave) radar sensors. This would solve the camera's vulnerability to poor weather
 - Migrating the processing pipeline from a standard Raspberry Pi 5 to a dedicated AI accelerator (such as the Google Coral Edge TPU or NVIDIA Jetson) to increase the frame rate.

Citations

- Yazdi, D. (2025, March 20). What Percentage of Car Crashes Are Caused by Driver Error? <https://accidentnetwork.com/what-percentage-of-car-crashes-are-caused-by-driver-error/>
- Szafranski, A. (2024, November 20). What is the 3 second rule for safe following distance? GoodCar. <https://goodcar.com/blog/3-second-rule-driving>
- Patuck, N. (2024, November 29). Deploying Ultralytics YOLO models on Raspberry Pi devices. Raspberry Pi. <https://www.raspberrypi.com/news/deploying-ultralytics-yolo-models-on-raspberry-pi-devices>